STA2023 Exam 2 — SIDE A — THE PRACTICE EXAM, SOLVED (the real exam copies it)

■ GREEN = the question ■ RED = rule / answer ■ BLACK = the work ■ BLUE = if they switch it up

Every part: formula → plug in → arithmetic → box it. Probabilities: 4 decimals (0.4 → 0.4000); μ, σ and x values: 2 decimals. No work = no credit.

Q1 (6 pts) — Two-way table: or / and / given

> **"The table below summarizes the employment status of 78 residents. Answer each question. Show your work."**

Status	Males	Females	Row total
Full-time	22	18	40
Part-time	6	10	16
Unemployed	3	2	5
Not in labor force	5	12	17
Column total	**36**	**42**	**78**

First: total each row and column at the table's edge (if the counts come in words, draw the table yourself). Every answer is then **a count ÷ the group you are picking from**.

(a) probability that a randomly selected resident is male? one category (a row or a column) → its total ÷ grand total (78)

$$P(\text{male}) = \frac{36}{78} = 0.4615$$

(b) …is part-time? same rule: add the row's cells, ÷ grand total

$$P(\text{part-time}) = \frac{6+10}{78} = \frac{16}{78} = 0.2051$$

(c) …is male OR unemployed? OR: $P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$ — adding counts the overlap (male AND unemployed = 3) twice, so take it out once

$$P(\text{M or U}) = \frac{36}{78} + \frac{5}{78} - \frac{3}{78} = \frac{38}{78} = 0.4872$$

Watch out: forgetting the overlap gives 41/78 × (check: 36 males + 2 unemployed women = 38 ✓)

(d) …is female AND not in the labor force (NLF)? AND in a table = the ONE cell where the row and column meet, ÷ grand total

$$P(\text{F and NLF}) = \frac{12}{78} = 0.1538$$

(e) …is part-time GIVEN that the resident is female? GIVEN: $P(B \mid A) = \frac{\#(A \text{ and } B)}{\#(A)}$ — the letter after | is the given group (the word after "given"): divide by its total, 42, not 78

$$P(\text{PT} \mid \text{Female}) = \frac{10}{42} = 0.2381$$

Watch out: 10/78 = 0.1282 is the classic error.

If they switch it up: NOT male → $1 - \frac{36}{78} = \frac{42}{78} = 0.5385$ · **female GIVEN part-time** → $\frac{10}{16} = 0.6250$ (direction matters) · **independent?** is P(PT | F) = P(PT)? 0.2381 ≠ 0.2051 → dependent (either direction gives the same verdict)

Q2 (2 pts) — Valid discrete probability distribution?

> **"Determine whether the following distribution represents a valid discrete probability distribution. If not, explain why." x: 0, 1, 2, 3, 4 with P(x): 0.34, 0.21, 0.13, 0.04, 0.01**

Valid only if (1) every $0 \le P(x) \le 1$ AND (2) $\sum P(x) = 1$. Write BOTH checks (given decimals must add to exactly 1).

Rule 1: 0.34, 0.21, 0.13, 0.04, 0.01 are all between 0 and 1 ✓

Rule 2: add them:

$$0.34 + 0.21 + 0.13 + 0.04 + 0.01 = 0.73 \neq 1 \times$$

NOT a valid probability distribution — the probabilities add to 0.73, not 1.

If they switch it up: any P(x) below 0 or above 1 → not valid even if the sum is 1 · sum above 1 (e.g. 1.02) → not valid · **"find the missing probability"** → 1 − (the others): if P(4) were blank, the rest add to 0.34 + 0.21 + 0.13 + 0.04 = 0.72 → P(4) = 1 − 0.72 = 0.28

Q3 (8 pts) — Stanley Cup: discrete random variable

> **"The following data represent the number of games played, X, in the Stanley Cup before a champion was determined." x: 4, 5, 6, 7 with frequency 20, 20, 26, 17.**

(a) construct the probability distribution

$P(x) = \frac{\text{frequency}}{\text{total}}$ **(frequency = how many times that x happened) total = 20 + 20 + 26 + 17 = 83**

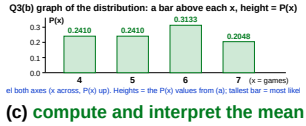
$$P(4) = \frac{20}{83} = 0.2410 \qquad P(5) = \frac{20}{83} = 0.2410$$

$$P(6) = \frac{26}{83} = 0.3133 \qquad P(7) = \frac{17}{83} = 0.2048$$

Check: 0.2410 + 0.2410 + 0.3133 + 0.2048 = 1.0001 ≈ 1 ✓ (rounding).

(b) graph the discrete probability distribution

x across, P(x) up; a bar above each x with height P(x); label both axes. In words: "bars at 4, 5, 6, 7 with heights 0.2410, 0.2410, 0.3133, 0.2048; tallest at x = 6."



(c) compute and interpret the mean of X

$\mu = \sum x \cdot P(x)$ **— a weighted average: each x counts as much as its probability**

$$\mu = 4(0.2410) + 5(0.2410) + 6(0.3133) + 7(0.2048)$$
$$= 0.9640 + 1.2050 + 1.8798 + 1.4336 = 5.48 \text{ games}$$

μ ≈ 5.48 games **Interpretation (part of the points):** over many Stanley Cup series, the average length is about 5.48 games.

(d) compute the standard deviation of X

$\sigma = \sqrt{\sum (x-\mu)^2 \cdot P(x)}$ **with μ = 5.48: subtract, square, × P(x), add, √**

x	x − μ	(x − μ)²	(x − μ)² · P(x)
4	−1.48	2.1904	2.1904(0.2410) = 0.5279
5	−0.48	0.2304	0.2304(0.2410) = 0.0555
6	0.52	0.2704	0.2704(0.3133) = 0.0847
7	1.52	2.3104	2.3104(0.2048) = 0.4732

$$\sigma^2 = 0.5279 + 0.0555 + 0.0847 + 0.4732 = 1.1413$$
$$\sigma = \sqrt{1.1413} = 1.07 \text{ games}$$

σ ≈ 1.07 games

Watch out: don't stop at the variance 1.1413; no ÷ 4 or ÷ 83 — the P(x) do the averaging. Write each product to 4 decimals and add those.

If they switch it up: P(at least 6 games) = P(6) + P(7) = 0.3133 + 0.2048 = 0.5181 · **P(fewer than 6)** = P(4) + P(5) = 0.4820 · **variance** = σ² = 1.1413 (stop before the √) · **expected value E(X)** = the mean

Q4 (8 pts) — Driving age: binomial

> **"According to a Gallup poll, 60% of U.S. women 18 years old or older stated that the minimum driving age should be 18. In a random sample of 10 U.S. women 18 years old or older, find the probability that:"**

Binomial: n = the sample size (10 yes/no trials); p = the percent in the question as a decimal (the thing being counted = "success"). **n = 10, p = 0.60, 1 − p = 0.40, X = number who say yes.**

$$P(x) = {}_nC_x\, p^x\, (1-p)^{n-x} \qquad {}_nC_x = \frac{n!}{x!\,(n-x)!}$$

(a) exactly 3 believe the minimum driving age should be 18

$${}_{10}C_3 = \frac{10!}{3!\,7!} = \frac{10 \cdot 9 \cdot 8}{3 \cdot 2 \cdot 1} = 120$$
$$P(3) = {}_{10}C_3\,(0.60)^3(0.40)^7 = 120(0.216)(0.0016384) = 0.0425$$

0.0425

Watch out: the power on 0.40 is n − x = 7, not 3. ${}_{10}C_3$: 10! = 10·9·8·7!, the 7! cancels → 10·9·8 ÷ (3·2·1), or press the nCr key.

(b) fewer than 3 believe… "fewer than 3" = 0, 1, 2 — the 3 is OUT. Add the three ($_{10}C_0$ = 1, 0.6^0 = 1); round each to 4 decimals, then add those.

$$P(0) = {}_{10}C_0(0.6)^0(0.4)^{10} = 1(1)(0.0001049) = 0.0001$$
$$P(1) = {}_{10}C_1(0.6)^1(0.4)^9 = 10(0.6)(0.0002621) = 0.0016$$
$$P(2) = {}_{10}C_2(0.6)^2(0.4)^8 = 45(0.36)(0.0006554) = 0.0106$$
$$P(X < 3) = 0.0001 + 0.0016 + 0.0106 = 0.0123$$

0.0123

(c) between 1 and 3 (inclusive) believe… inclusive → x = 1, 2, 3. Reuse (a) and (b) — write "from part (a)".

$$P(1 \le X \le 3) = P(1) + P(2) + P(3) = 0.0016 + 0.0106 + 0.0425 = 0.0547$$

0.0547

(d) sample of 200: expected number and standard deviation binomial shortcuts: $\mu = np$, $\sigma = \sqrt{np(1-p)}$

$$\mu = np = 200(0.60) = 120$$
$$\sigma = \sqrt{np(1-p)} = \sqrt{200(0.60)(0.40)} = \sqrt{48} = 6.93$$

μ = 120, σ ≈ 6.93 (48 is the variance — take the √)

If they switch it up: at least 3 = 1 − P(X < 3) = 1 − 0.0123 = 0.9877 · **at most 3** (3 is IN) = P(X < 3) + P(3) = 0.0123 + 0.0425 = 0.0548 · **more than 3** = 1 − 0.0548 = 0.9452 · **at least 1** = 1 − P(0) = 1 − $(0.4)^{10}$ = 0.9999 · **"would 130 of 200 be unusual?"** np(1 − p) = 48 ≥ 10 ✓ so the μ ± 2σ rule applies: 120 ± 2(6.93) = 106.1 to 133.9 → 130 inside → not unusual

Q5 (5 pts) — Daily temperature: uniform

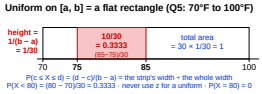
> **"The daytime temperature X (°F) in a coastal city during summer is uniformly distributed between 70°F and 100°F."**

a = smaller end 70, b = bigger end 100, width 30. $\text{height} = \frac{1}{b-a}$ $P(c \le X \le d) = \frac{d-c}{b-a}$ **= the piece's share of the width. "more than c" → (b − c)/(b − a); "less than d" → (d − a)/(b − a). No z (that's Q6).**

(a) sketch / describe the pdf (= the probability graph)

$$\text{height} = \frac{1}{b-a} = \frac{1}{100-70} = \frac{1}{30} \qquad \text{area} = 30 \cdot \frac{1}{30} = 1 \checkmark$$

a flat line (rectangle) at height 1/30 from 70 to 100, and 0 everywhere else



(b) P(75 ≤ X ≤ 85)

$$P(75 \le X \le 85) = \frac{85-75}{100-70} = \frac{10}{30} = 0.3333$$

0.3333

(c) P(X < 80)

$$P(X < 80) = \frac{80-70}{100-70} = \frac{10}{30} = 0.3333$$

0.3333

Watch out: P(X ≤ 80) is the same number — one point has no width, P(X = 80) = 0.

If they switch it up: P(X > 90) = $\frac{100-90}{30} = 0.3333$ · **P(X ≥ 95)** = 5/30 = 0.1667 · **"20% of days are cooler than what temperature?"** → the 20% goes in as 0.20: $x = a + 0.20(b-a) = 70 + 0.20(30) = 76°\text{F}$ (no z)

Q6 (6 pts) — Standard normal: the exam's z hint (= the table entry) is the area to the LEFT

> **"Use the standard normal table (on the real exam: the printed z hints). Round final answers to four decimal places."**

No table on this sheet: the exam prints the values as hints. LEFT (below) = the hint · RIGHT (above) = 1 − hint · BETWEEN = (left of the bigger z) − (left of the smaller z); bigger = further right, so −0.25 is bigger than −1.30 · $z = \frac{x-\mu}{\sigma}$ **to 2 decimals**

(a) area to the left of z = −1.10

the hint (or table row −1.1, column .00): **0.1357**

Watch out: a hint P(Z < −1.10) = 0.1357 is already the left area — use it as is (only with no negative z given: 1 − 0.8643 = 0.1357).

(b) area between z = −0.60 and z = 1.45

hints: left of 1.45 = 0.9265; left of −0.60 = 0.2743.

$$\text{between} = 0.9265 - 0.2743 = 0.6522$$

0.6522

Watch out: subtract the AREAS, never the z's; a negative answer means you subtracted the wrong way.

(c) μ = 80, σ = 10: P(70 < X < 95)

$$z_1 = \frac{70-80}{10} = -1.00 \;\rightarrow\; \text{left} = 0.1587$$
$$z_2 = \frac{95-80}{10} = 1.50 \;\rightarrow\; \text{left} = 0.9332$$
$$P(70 < X < 95) = 0.9332 - 0.1587 = 0.7745$$

0.7745

If they switch it up: right of z = 1.45 = 1 − 0.9265 = 0.0735 · **P(X > 95)** = 1 − 0.9332 = 0.0668 · **z with area 0.10 to its LEFT** → closest entry 0.1003 → z = −1.28 · **z with area 0.025 to its RIGHT** → left area 0.975 → z = 1.96

Q7 (8 pts) — Wechsler IQ: normal with μ = 100, σ = 15

> **"The Wechsler Intelligence Scale for Children is approximately normally distributed with mean μ = 100 and standard deviation σ = 15."**

Forward (a)–(c): $z = \frac{x-100}{15}$ **→ hint → below: left · above: 1 − left · between: bigger left − smaller left. Backward (d): percent → left area → the hint's z →** $x = \mu + z\sigma$ **(x to 1 decimal; 130.75 ≈ 131 is fine). Unusual: |z| > 2 or tail area < 0.05 — no hint needed.**

(a) probability a test taker scores above 125

$$z = \frac{125-100}{15} = 1.67 \;\rightarrow\; \text{left} = 0.9525$$
$$P(X > 125) = 1 - 0.9525 = 0.0475$$

0.0475

Watch out: 0.9525 is the #1 error; fewer than half score above 125 → the answer must be below 0.5.

(b) probability a test taker scores below 90

$$z = \frac{90-100}{15} = -0.67$$
$$P(X < 90) = \text{left of } -0.67 = 0.2514$$

0.2514 (keep the minus sign on z)

(c) proportion scoring between 110 and 140

$$z_1 = \frac{110-100}{15} = 0.67 \;\rightarrow\; \text{left} = 0.7486$$
$$z_2 = \frac{140-100}{15} = 2.67 \;\rightarrow\; \text{left} = 0.9962$$
$$P(110 < X < 140) = 0.9962 - 0.7486 = 0.2476$$

0.2476 (proportion = probability; 4 decimals)

(d) score at the 98th percentile

98th percentile = 98% score below it → area to the LEFT = 0.98 → the exam's hint gives z = 2.05 (from a table: 0.9798 is the entry closest to 0.98).

$$x = \mu + z\sigma = 100 + 2.05(15)$$
$$= 100 + 30.75 = 130.75 \approx 131$$

≈ 131 (130.75)

Watch out: never put 0.98 into the z formula — it is an area, not a score.

If they switch it up: top 5% → left 0.95 → z = 1.645 (halfway between 1.64 and 1.65) → 100 + 1.645(15) = 124.7 · **bottom 10%** → left 0.10 → z = −1.28 → 100 − 1.28(15) = 80.8 · **middle 95%** → 0.025 in each tail → z = ±1.96 → 70.6 to 129.4 · **is 70 unusual?** z = −2.00 → left area 0.0228 < 0.05 → yes (|z| > 2 also says unusual)

STA2023 Exam 2 — SIDE B — RULES, FORMULAS, PICTURES

■ BLUE = what it means ■ RED = formula / rule ■ GREEN = example ■ BLACK = steps

SYMBOLS — what they mean

Symbol	Plain English
Σ	"add them all up"
N / n	how many values (N population, n sample)
μ ("mu") / x̄ ("x-bar")	the mean (average): μ population, x̄ sample
σ ("sigma") / s	standard deviation (SD) = typical gap from the mean; σ², s² = variance = SD²
z	z-score = how many SDs a value sits above (+) or below (−) the mean
P(E), E^c, N(E)	probability of event E (0 = impossible … 1 = certain); E^c = "not E"; N(E) = # of outcomes in E, N(S) = # of all outcomes
P(F | E)	probability of F **given** you already know E happened
X / x, P(x)	X = the random variable (the number an experiment produces); x = one value; P(x) = the chance X comes out equal to x; μ_X, σ_X = its mean and SD; E(X) = expected value = μ_X
n, p, 1 − p; ${}_nC_r$, ${}_nP_r$, n!	binomial: # of trials, P(success), P(failure); C = pick r of n, order ignored; P = order counts; n! = n·(n−1)·… ·1

μ, σ or capital N → whole population; x̄, s or small n → a sample. ≤ / ≥ = at most / at least (the end number counts); ≠ = not equal. Probabilities: 4 decimals (0.4 → 0.4000); μ, σ, x: 2.

WHICH TOOL? — read the question first (★Qn = solved on side A)

Question says…	Use
table of counts + "or / and / given"	Ch 4 rules on the two-way table **★Q1**
"how many ways"	counting: multiply / ${}_nP_r$ / ${}_nC_r$
x with its P(x), or x with its frequency count	Ch 5: validity check, μ, σ **★Q2 ★Q3**
n trials, probability p, "how many successes"	binomial; μ = np, σ = √(np(1−p)) **★Q4**
"uniformly distributed between a and b"	uniform: (d − c)/(b − a), never z **★Q5**
"normally distributed, mean μ, SD σ"	z → table → left / right / between **★Q6 ★Q7**
"what value is the __th percentile / top 10%"	backwards: area → z → x = μ + zσ **★Q7(d)**

EXAM 1 SKILLS THAT COME BACK

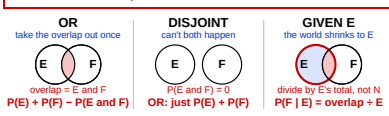
- **Mean** $\mu = \frac{\sum x}{N}$ (balance point); **median** = middle of the sorted list; **mode** = most frequent. **SD** = typical distance from the mean: $\sigma = \sqrt{\frac{\sum(x-\mu)^2}{N}}$ (sample s: ÷ (n − 1); calculator σx = population, Sx = sample).
- **z-score** $z = \frac{x-\mu}{\sigma}$ = how many SDs above (+) or below (−) the mean; beyond ±2 → unusual. **Empirical Rule** 68–95–99.7 (picture in the normal section). **Quartiles: Q_1, Q_3 = the 25th, 75th percentile values; IQR = $Q_3 - Q_1$; outlier if below $Q_1 - 1.5\cdot IQR$ or above $Q_3 + 1.5\cdot IQR$**

CH 4 — PROBABILITY

- **Probability** = the long-run fraction of times something happens (Law of Large Numbers: more trials → the fraction settles at the probability). **0 ≤ P(E) ≤ 1; all outcomes together add to 1. Unusual event** = probability below 0.05.
- **Sample space S** = everything that could happen; an **event** = any part of it.
- **Classical** (equally likely outcomes — dice, cards, coins): count. $P(E) = \frac{N(E)}{N(S)}$ P(two heads) = 1/4; P(at least one tail) = 3/4.
- **Empirical** (from data): $P(E) \approx \frac{\text{frequency of } E}{\text{number of trials}}$ 22 of 200 carpool → 0.11 (interpret: "if many people were selected, about 11 of every 100 would carpool"); 5 walk → 5/200 = 0.025 < 0.05 → unusual. **Subjective** = educated guess.
- **Disjoint** (mutually exclusive) = can't both happen (one card: king AND queen), so P(E and F) = 0. **Independent** = E happening doesn't change P(F) (two coin flips). **Disjoint is NOT independent:** if they are disjoint, knowing E happened means F didn't.

The rules — spot the keyword

NOT: $P(E^c) = 1 - P(E)$ · **AT LEAST ONE:** $P(\text{at least 1}) = 1 - P(\text{none})$
OR: $P(E \text{ or } F) = P(E) + P(F) - P(E \text{ and } F)$ **(disjoint → the overlap is 0, just add)**
AND, independent: $P(E \text{ and } F) = P(E)\cdot P(F)$ · **AND, always:** $P(E \text{ and } F) = P(E)\cdot P(F \mid E)$
GIVEN: $P(F \mid E) = \frac{P(E \text{ and } F)}{P(E)} = \frac{N(E \text{ and } F)}{N(E)}$.
Independent? check $P(F \mid E) = P(F)$: equal → independent, different → dependent



Cards: 52 = 4 suits × 13 ranks (4 per rank, 13 per suit, 12 face cards J/Q/K). P(king or ♦) = 4/52 + 13/52 − 1/52 = 16/52 = **0.3077**. Two picks, no replacement: P(2 kings) = 4/52 · 3/51 = **0.0045** (the 1st card stays out, so the 2nd pick is 3 kings of 51 cards).

- **"At least one":** compute the one case you don't want (none) and subtract from 1. P(survive) = 0.9986: P(both of 2 survive) = 0.9986^2 = 0.9972; P(at least one of 1000 dies) = $1 - 0.9986^{1000}$ = 0.7536.

Counting — "how many ways"

- A **sequence of choices**: multiply the options at each step. 3-letter codes: 26·26·26 = 17,576; chair, vice-chair, secretary from 14: 14·13·12 = 2,184.
- **Permutation** = pick r of n, ORDER MATTERS (1st/2nd/3rd). ${}_nP_r = \frac{n!}{(n-r)!}$ ${}_7P_5 = 7\cdot6\cdot5\cdot4\cdot3 = 2{,}520$ (r numbers counting down); ${}_{10}P_3 = 720$; 7 in order: 7! = 5,040; note 0! = 1: ${}_5P_5 = 5! = 120$; ${}_nC_0 = {}_nC_n = 1$.
- **Combination** = pick r of n, order DOESN'T matter (committee, poker hand, sample). ${}_nC_r = \frac{n!}{r!(n-r)!}$ ${}_4C_2 = 4!/(2!\cdot2!) = 24/(2\cdot2) = 6$; ${}_{20}C_4 = 4{,}845$. Probability by counting: P(4 cards all the same rank) = 13 ranks ÷ ${}_{52}C_4$ = 13/270,725 ≈ 0.00005.

CH 5 — DISCRETE RANDOM VARIABLES ★Q2 ★Q3

- **Random variable X** = the number that comes out of a chance experiment. **Discrete** = you count it; its possible values are a list (x = 0, 1, 2, …). **Continuous** = you measure it; its possible values are an interval (time t ≥ 0, mass 0 < m ≤ 16 oz). **Probability distribution** = every value X can take with its probability.

VALID only if (1) every $0 \le P(x) \le 1$ AND (2) $\sum P(x) = 1$
$\mu_X = E(X) = \sum [x \cdot P(x)]$
$\sigma_X = \sqrt{\sum [(x-\mu)^2 \cdot P(x)]}$

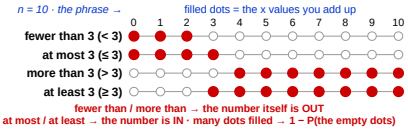
Mean of a distribution = weighted average (each value counts as much as its probability) = the long-run average per trial. **SD** = typical distance from the mean, each squared gap weighted by its probability.

Q: "Insurance: premium $100, pays $10,000 if the client dies, P(dies) = 0.0014. Expected profit?" **A:** build X = profit from the words: lives → +$100, P = 1 − 0.0014 = 0.9986; dies → 100 − 10,000 = −$9,900, P = 0.0014 (sum = 1 ✓). E(X) = 100(0.9986) − 9900(0.0014) = 99.86 − 13.86 = **$86** per policy (positive → keep selling).

CH 5 (cont.) — BINOMIAL ★Q4

Binomial experiment = the same yes/no trial repeated: (1) FIXED number of trials n, (2) independent, (3) two outcomes (success / failure), (4) SAME p each time. X = number of successes. Not binomial if n isn't fixed ("until…") or picks without replacement are more than 5% of the group.

$P(x) = {}_nC_x\, p^x (1-p)^{n-x}$ $\quad {}_nC_x = \frac{n!}{x!(n-x)!}$
$\mu = np$ $\quad \sigma = \sqrt{np(1-p)}$
Unusual count (needs np(1 − p) ≥ 10): below μ − 2σ or above μ + 2σ · Excel: BINOM.DIST(x, n, p, FALSE) = P(x); TRUE = P(X ≤ x)



Q: "Binomial or not? (1) Roll a pair of dice 10 times, count sums of 7. (2) Select flights until 10 late ones. (3) Pick 4 of 30 students, count females." **A:** (1) YES: n = 10, p = 6/36 (6 of the 36 sums are 7), x = 0, 1, …, 10 (x always runs 0 … n). (2) NO: n not fixed. (3) NO: no replacement, and 4 of 30 = 13% > 5% of the group → p changes.

CH 6 — CONTINUOUS & NORMAL

- A continuous variable has infinitely many values, so probability = **AREA under the curve** (pdf = the probability curve: never below 0, total area 1). **P(X = one exact value) = 0** so P(a < X < b) = P(a ≤ X ≤ b).

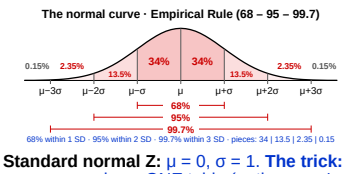
Uniform ★Q5

Uniform from a to b = a flat rectangle: every stretch of the same length is equally likely. Probability of an interval = its share of the total width.

$\text{height} = \frac{1}{b-a}$ $\quad P(c \le X \le d) = \frac{d-c}{b-a}$
value with share P below it: $x = a + P(b-a)$

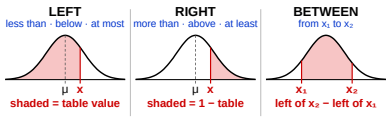
Normal ★Q6 ★Q7

Normal = the bell curve: symmetric around μ, most values near the middle. μ slides it left/right, σ makes it wider or narrower. **Properties:** symmetric about μ; mean = median = mode = μ (the peak); bends (inflection points) at μ ± σ; total area 1; tails never touch the axis; Empirical Rule; half the area on each side of μ.



Standard normal Z: μ = 0, σ = 1. **The trick:** convert any value to a z-score and use ONE table (or the exam's hint): it gives the area to the LEFT of z.

$z = \frac{x-\mu}{\sigma}$ (round to 2 decimals) $\quad x = \mu + z\sigma$
LEFT ("less than / below"): table value · RIGHT ("more than / above"): 1 − table value
BETWEEN: (left of the bigger z) − (left of the smaller z) · no −z row in the table? left of −z = 1 − (left of +z)



Steps: draw the bell, mark μ and x, shade → z = (x − μ)/σ to 2 decimals → hint or table (z = 1.67: row 1.6, column .07) = area to the LEFT → left: done · right: 1 − it · between: big left − small left.

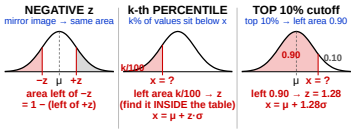
USING THE z HINTS ON THE EXAM (no table needed)

Dr. Amir: every z-score you need is printed as a hint on the exam paper; a hint is a table value, so use it exactly like the table. Still compute your own z = (x − μ)/σ to 2 decimals and match it to a hint — no match means your arithmetic is wrong.

"P(Z < 1.67) = 0.9525" / "area to the left of 1.67 is 0.9525" → the LEFT area: below → write it · above → 1 − it · between → bigger − smaller
"P(Z > 1.67) = 0.0475" (a right tail) → that IS the "above" answer; "below" = 1 − 0.0475
"z = 2.05 for the 98th percentile" / "z = 1.28 for the top 10%" → x = μ + z·σ; bottom 10% → negative z (z = −1.28 → x = μ − 1.28σ)
"P(Z < −1.43) = 0.0764" is already a LEFT area — write it as is · "is it unusual?" has no hint: |z| > 2 (x beyond μ ± 2σ) → unusual

Backwards — find x from a percent / percentile ★Q7(d)

The area is GIVEN; run the steps backwards: words → area to the LEFT ("top 10%" → 0.90, "bottom 5%" → 0.05, "k-th percentile" → k/100, "middle 80%" → 0.10 and 0.90) → find it INSIDE the table (closest entry; 0.95 sits halfway between 1.64 and 1.65 → 1.645; 0.995 → 2.575) → read z from its row + column → x = μ + zσ.



Percentile → z: 1st −2.33 · 5th −1.645 · 10th −1.28 · 20th −0.84 · 25th −0.67 · 75th 0.67 · 80th 0.84 · 90th 1.28 · 95th 1.645 · 97.5th 1.96 · 98th 2.05 · 99th 2.33 · middle 80 / 90 / 95 / 99% ↔ ±1.28 / ±1.645 / ±1.96 / ±2.575

- **"Give two interpretations" of an area:** "the area to the right of 200 is 0.2903" means (1) the **proportion** of the population above 200 is 0.2903 (29.03%) AND (2) the **probability** that one randomly selected person is above 200 is 0.2903.
- **Is it normal?** histogram bell-shaped, one peak, symmetric, few outliers → yes; a **normal probability plot** (observed values vs the z-scores expected if normal) close to a straight line → yes; a long tail or a bent plot → not normal.

DON'T LOSE POINTS

- "Or" → subtract the overlap once. "And" → the one table cell. "Given" → divide by the given group, not the grand total. Σ P(x) = 1 and each P(x) 0–1 before calling a distribution valid.
- Binomial counts: "fewer than 3" = 0, 1, 2; "at least 3" = 1 − P(X ≤ 2); "at most 3" = 0, 1, 2, 3. Binomial σ needs the √. Uniform: length ÷ total width, never z.
- Normal: z to 2 decimals; the hint or table value = area to the LEFT; "above" → 1 − left; "between" → big − small. Percentile → area → z (inside the table) → x = μ + zσ. 4 decimals; formula every time.

WRITE THIS SENTENCE — answer templates

- **Interpret the mean / expected value:** "Over many [series], the average [games per series] is about [5.48]." Insurance: "$86 profit per policy on average → positive → keep selling."
- **Valid distribution? (Q2):** "Not valid: the probabilities add to 0.73, not 1" / "not valid: P(2) = −0.1 is negative" / "each P(x) is 0–1 and they add to 1 → valid."
- **Binomial or not:** "Yes: fixed n = 10, independent, two outcomes, same p = 0.60; x = 0, 1, …, 10." / "No: n not fixed." / "No: no replacement; the sample is 4/30 = 13% > 5% of the group, so p changes."
- **Two interpretations of an area** (0.2903 right of 200): "(1) 29.03% of males are above 200. (2) The probability a randomly selected male is above 200 is 0.2903."
- **Unusual?** "P = 0.025 < 0.05 → unusual" / "count 130 is inside μ ± 2σ = 106.1 to 133.9 → not unusual." **Independent?** "P(A | B) = 0.2381 ≠ P(A) = 0.2051 → dependent." **Sketch:** uniform = "flat line at height 1/30 from 70 to 100, 0 elsewhere, area 1"; normal = bell at μ, ticks μ ± σ, 2σ, 3σ, shade, write the area.

Calculator: 36 ÷ 78 = 0.4615 · 0.4 ^ 7 = 0.0016384 · 10 [nCr] 3 = 120 (TI-84: MATH → PRB; TI-89: 2nd MATH → Probability → nCr(10,3); no key: 10! ÷ (3!·7!)) · √48 = 6.93 · keep decimals (round only z, to 2).
Order (minutes): Q1 6 · Q2 2 · Q5 5 · Q6 6 · Q7 9 · Q3 10 · Q4 10. Formula first, box every answer.